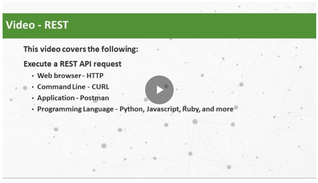
REST

TASK 1

Video - REST

As you have just learned, REST is currently the most widely used API. This topic covers REST in more detail.

Click play in the video to learn about more about REST.



TASK 2

REST and RESTful API

Web browsers use HTTP or HTTPS to request (GET) a web page. If successfully requested (HTTP status code 200), web servers respond to GET requests with an HTML coded web page, as shown in the figure.

REST is an architectural style for designing web service applications. It refers to a style of web architecture that too many underlying characteristics and governs the behavior of clients and servers. Simply stated, a REST API is an API that works on top of the HTTP protocol. It defines a set of functions developers can use to perform requests and receive responses via HTTP protocol such as GET and POST.

Conforming to the constraints of the REST architecture is generally referred to as being "RESTful". An API can be considered "RESTful" if it has the following features:

- **Client-Server** – The client handles the front end and the server handles the back end. Either can be replaced independently of the other.
- **Stateless** – No client data is stored on the server between requests. The session state is stored on the client.
- **Cacheable** – Clients can cache responses to improve performance.

TASK 3

RESTful Implementation

A RESTful web service is implemented using HTTP. It is a collection of resources with four defined aspects:

- The base Uniform Resource Identifier (URI) for the web service, such as http://example.com/resources/.
- The data format supported by the web service. This is often JSON, YAML, or XML but could be any other data format that is a valid hypertext standard.
- The set of operations supported by the web service using HTTP methods.
- The API must be hypertext driven.

RESTful APIs use common HTTP methods including POST, GET, PUT, PATCH and DELETE. As shown in the following table, these correspond to RESTful operations: Create, Read, Update, and Delete (or CRUD).

HTTP Method	RESTful Operation
POST	Create
GET	Read
PUT/PATCH	Update
DELETE	Delete

In the figure, the HTTP request asks for JSON-formatted data. If the request is successfully constructed according to the API documentation, the server will respond with JSON data. This JSON data can be used by a client's web application to display the data. For example, a smartphone mapping app show the location of San Jose, California.

TASK 4

URI, URN, and URL

Web resources and web services such as RESTful APIs are identified using a URI. A URI is a string of characters that identifies a specific network resource. As shown in the figure, a URI has two specializations:

- **Uniform Resource Name (URN)** - Identifies only the namespace of the resource (web page, document, image, etc.) without reference to the protocol.
- **Uniform Resource Locator (URL)** - defines the network location of a specific resource on the network. HTTP or HTTPS URLs are typically used with web browsers. Other protocols such as FTP, SFTP, SSH, and others can use a URL. A URL using SFTP might look like: sftp://city.example.com

These are the parts of a URI, as shown in the figure:

- **Protocol/scheme** – HTTPS or other protocols such as FTP, SFTP, mailto, and NNTP
- **Hostname** – www.example.com
- **Path and file name** – /author/book.html
- **Fragment** – #page155

Parts of a URI

TASK 5

Anatomy of a RESTful Request

In a RESTful Web service, a request made to a resource's URI will elicit a response. The response will be a payload typically formatted in JSON, but could be HTML, XML, or some other format. The figure shows the URI for the MapQuest directions API. The API request is for directions from San Jose, California to Monterey, California.

Parts of an API Request

The figure shows part of the API response. In this example it is the MapQuest directions from San Jose to Monterey in JSON format.

Partial JSON Payload Received from an API Request


```
{
  - route: {
      hasTollRoad: false,
      hasBridge: true,
      - boundingBox: {
          - lr: {
              lng: -121.667861,
              lat: 36.594869
            },
          - ul: {
              lng: -122.897125,
              lat: 37.335350
            }
        },
      distance: 71.712,
      hasTimedRestriction: false,
      hasTunnel: false,
      hasHighway: true,
      computedWaypoints: [ ],
      - routeError: {
          errorCode: -400,
          message: ""
        },
      formattedTime: "01:12:59",
      sessionId: "5ce1ebf7-017f-5f21-02b4-1a7b-04b0810df024",
      hasAccessRestriction: false,
      realTime: 4379,
      hasSeasonalClosure: false,
      hasCountryCross: false,
      fuelUsed: 3.29,
      - legs: {
          - {
              hasTollRoad: false,
              hasBridge: true,
              destNarrative: "Proceed to MONTEREY, CA.",
              distance: 71.712,
```


These are the different parts of the API request:

- **API Server** – This is the URL for the server that answers REST requests. In this example it is the MapQuest API server.
- **Resources** – Identifies the data format and information the client is requesting from the API service. Queries can include:
  - **Format** – This is usually JSON but can be YAML or XML. In this example JSON is requested.
  - **Key** – The key is for authorization, if required. MapQuest requires a key for their directions API. In the above URL you would need to replace "KEY" with a valid key to submit a valid request.
  - **Parameters** – Parameters are used to send information pertaining to the request. In this example, the query parameters include information about the directions that the API needs so it knows what directions to return: "from=San+Jose,Ca" and "to=Monterey,Ca".

Many RESTful APIs, including public APIs, require a key. The key is used to identify the source of the request. Here are some reasons why an API provider may require a key:

- To authenticate the source to make sure they are authorized to use the API.
- To limit the number of people using the API.
- To limit the number of requests per user.
- To prevent malicious use of the API by requested by users.
- To gather information on the people using the API.

Note: If you wish to use the MapQuest API, the API does require a key. Search the Internet for the URL to obtain a MapQuest key. Use the search parameters: developer.mapquest. You can also search the Internet for the current URL that outlines the MapQuest privacy policy.

TASK 6

RESTful API Applications

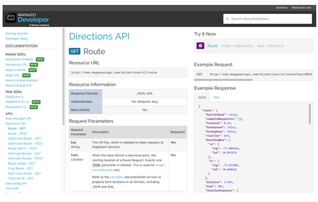
Many web sites and applications use APIs to access information and provide service for their customers. For example, when using a travel service web site, the travel service uses the API of various airlines to provide the user with airline, hotel and other information.

Some RESTful API requests can be made by typing in the URI from within a web browser. The MapQuest directions API is an example of this. A RESTful API request can also be made in other ways.

Click each API application scenario below for more information.

Developer Web Site | Postman | Python | Network Operating Systems

Developers often maintain web sites that include information about the API, parameter information, and usage examples. These sites may also allow the user to perform the API request within the developer web page by entering in the parameters and other information. The following figure shows an example of the MapQuest Directions API web page.



TASK 7

Check Your Understanding - REST

Check your understanding of REST by choosing the BEST answer to the following questions.

1. An API in considered RESTful if it has which of the following features? (Choose three.)

- [ ] Stateful
- [x] Stateless
- [x] Cacheable
- [ ] Cacheless
- [x] Client-Server
- [ ] Server-Server

2. Which of the following is a URL?

- https://www.cisco.com/c/en/us/solutions/design-zone.html#-all-guides
- https://www.cisco.com/c/en/us/solutions/design-zone.html
- www.cisco.com/c/en/us/solutions/design-zone.html
- #-all-guides

3. Which of the following is a URN?

- https://www.cisco.com/c/en/us/solutions/design-zone.html#-all-guides
- https://www.cisco.com/c/en/us/solutions/design-zone.html
- www.cisco.com/c/en/us/solutions/design-zone.html
- #-all-guides

4. Which of the following is a URI?

- https://www.cisco.com/c/en/us/solutions/design-zone.html#-all-guides
- https://www.cisco.com/c/en/us/solutions/design-zone.html
- www.cisco.com/c/en/us/solutions/design-zone.html
- #-all-guides

5. Which of the following make up the query portion of a RESTful request? (Choose three.)

- [x] Key
- [x] Format
- [x] Parameters
- [ ] Resources
- [ ] API server

Check | Show Me | Reset